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Prevalence and predictors of tobacco use among Saudi adolescents: the Global Youth Tobacco Survey (GYTS) 2022

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Abstract

Background Adolescents are particularly vulnerable to acquiring new behaviors, including tobacco use, especially in familiar and social gatherings and celebrations where they witness certain behaviors like waterpipe use. Saudi Arabia enhances preventive and therapeutic healthcare services. This study aimed to determine the prevalence of tobacco use and identify its predictors among adolescents aged 13–15 years in Saudi Arabia.

Methods Data were collected from the Global Youth Tobacco Survey, which was conducted in Saudi Arabia in 2022.

Results The overall prevalence of tobacco use among Saudi Arabian adolescents was 8.8%, where 7.5% of the participant females versus 10% of the males use tobacco products. Age and lower spending were significantly associated with higher evidence of smoking among youth ($P < 0.05$). Moreover, various predictors ofsmoking behavior among Saudi adolescents e.g., exposure to secondhand smoke, elevate the likelihood of the adolescents in Saudi Arabia to use tobacco (OR 1.849, CI 1.450–2.358). Additionally, different kinds of media messages target this age category as well as the adolescent's perception of smoking (OR 1.319, CI 1.021–1.705).

Conclusion Increased efforts are needed to address the harmful health and social impacts of smoking among adolescents in Saudi Arabia. Such efforts may include educational health campaigns targeting adolescents at their schools or in public places and mass gatherings.

Keywords Prevalence, Predictors, Adolescents, Tobacco use, Smoking, Saudi Arabia

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Background

Tobacco smoking is a fundamental public health issue and is responsible for over eight million deaths annually [1]. In the last two decades, There has been increased exposure of young people to tobacco marketing and advertising during their teenage years, thereby facilitating initiation [2]. Early onset smoking among adolescents presents immediate risks to their health, initiates chronic addiction, and interferes with the developing brains of young people from nicotine use [3]. Evidence showed that early initiation of smoking increases the risk for progression to daily smoking, increased use, sustained nicotine dependence, and worse difficulty quitting [4]. Although in the last two decades there has been a decrease in tobacco smoking worldwide, findings suggest that 88% of adult smokers had their first experience with a cigarette at or under the age of 18 [5, 6] Unfortunately, trends indicate a rapid increase in tobacco use among individuals aged 11–15 years, particularly by the age of 15, in both genders [5, 7]. Moreover, European, United States, African, and Asian reports indicate a rise in the consumption of e-cigarettes among teenagers, even surpassing adult consumption in some cases [8]. These reports even show that adolescent e-cigarette use outnumbers adult use [9]. Additionally, studies have reported the average rate of tobacco usage among youths in the Middle East is relatively high compared with that in other regions [6, 10–12]. It was Also, reported by the study of [9] that the Eastern Mediterranean region has the greatest global prevalence of non-cigarette tobacco usage (16.7% in boys and 9.0% in girls), surpassing that of cigarette smoking (12.2% in males and 3.4% in girls). Another study in United Arab Emirates indicated that a total of 1047 (17%) participants reported having ever smoked cigarettes or tobacco, while 818 (14%) were current smokers. The same study revealed that 24% of male adolescents reported current smoking compared to 5.5% among females [12]. Additionally, the Eastern Mediterranean Region Adolescent Health Collaborators stated that Daily tobacco smoking rates among males aged 10–24 years varied from 1.9% in Sudan to 18% in Kuwait, but were less than 5% for females in the region [13]. On the other hand, Dalibalta and others (2023) reported that traditional cigarette use is declining, the market for alternative tobacco products (ATPs), such as electronic cigarettes, smokeless tobacco, pipes, and hookah, is growing rapidly in the Gulf region [8]. According to Shubayr and colleagues, (2024) tobacco consumption rates vary globally, with some countries experiencing significant reductions due to health programs and legislation, while others, including developing countries and high-income nations like Saudi Arabia, have seen concerning increases [14]. Also, Ma et al., stated that the Eastern Mediterranean region is one of only two regions that did not reduce the use of cigarettes over a decade (the other being South-East Asia). Additionally, the region witnessed a substantial rise in non-cigarette tobacco use (3.5% over 10 years, primarily among boys). In Saudi Arabia, tobacco smoking has become a significant public health challenge, particularly among youth. The increasing smoking rates in this group are alarming because of associated short- and long-term health risks. According to the results of the GYTS [15], approximately 17.7% of students aged 13–15 years have smoked tobacco [5]. In addition to the conventional cigarettes and waterpipes, another study reported that e-cigarettes are sold predominantly in online marketplaces, supermarkets, gas stations, and shopping centers, making these the most common commercial sources of e-cigarettes in Saudi Arabia [16]. Also, social sources of e-cigarettes, such as obtaining them from friends, have also been reported [16]. A systematic review in Saudi Arabia identified smoking among friends and family members as a major risk factor for adolescent smoking. In addition, common reasons for waterpipe smoking include being dissatisfied, having free time, socializing with friends, and being bored [6]. Also, a recent study in Saudi Arabia reported that adolescents are using e-cigarettes at unsettling rates [17]. The problem of adolescent smoking is sophisticated due to the numerous interrelated factors influencing adolescent smoking such as social and physical environments, which complicates the issue. For instance, the portrayal of tobacco as a normative behavior in media significantly encourages its use [18]. Also, adolescents who observe their peers or parents using tobacco, and those who perform poorly in school, or experience anxiety, depression, stress, or low self-esteem are more likely to smoke [15]. Therefore, the World Health Organization (WHO) has identified adolescent smoking as one of the most critical global public health threats [15]. To overcome the challenge of increasing tobacco use among youth at both the national and global levels, the WHO established the Framework Convention on Tobacco Control (FCTC) in 2003 to help countries address tobacco consumption. The FCTC employs evidence-based initiatives to reduce the supply and demand for tobacco through behavioral and environmental changes [19]. This framework identified six tobacco control measures, known as MPOWER, that aims at reducing global tobacco use. These measures include monitoring tobacco use and tobacco control measures, providing cessation programs, warning people about the harmful effects of smoking (e.g., through warning labels and media campaigns), banning tobacco advertising and promotions, and raising taxes on tobacco products [19]. By October 2020, 168 countries signed the WHO FCTCz, and 182 had ratified it including the GCC countries have ratified the World Health Organization (WHO) (FCTC) [20]. As part of Saudi Vision 2030's

emphasis on human health and well-being, particularly among improving public health and preventing diseases. Addressing smoking behavior and its associated health impacts is a significant priority [21]. In 2005 Saudi Arabia has ratified and become a party to the WHO FCTC. Since that the Saudi Arabia's national anti-smoking council is striving toward more intense tobacco regulation and established an anti-smoking law "Anti-Smoking Law Royal Decree No. (M/56)" in 2015 to prevent tobacco usage [22] and its implementation regulations in 2019 [23]. This law aims to implement all required steps at the national, community, and personal levels to reduce the smoking habit among people of all ages. Concerning adolescents, these measures include restrictions on selling tobacco products for those under 18 years age, prohibiting tobacco promotion and advertising, and increasing taxes and fines on tobacco products. Moreover, the law imposed establishing tobacco free places (public, and private) [22]. Many initiatives were developed by Saudi Arabia to combat smoking e.g., antismoking campaigns to increase public awareness, smoking clinics to help those intending to quit and provide the required medications and mobile application to detect tobacco prohibition violations.

Given the need for continuous monitoring and evaluation of the predictors and trends especially considering the strength of the tobacco business and the multifaceted nature of tobacco use, this research leverages GYTS data (2022) from the Saudi Ministry of Health's tobacco control program. The goal is to investigate the prevalence to identify the predictors and to explore factors influencing accessibility to tobacco products among adolescents aged 13–15 years in Saudi Arabia. Additionally, to develop recommendations for effective prevention and intervention strategies.

Methods

Aim

This study investigated the prevalence and predictors of tobacco use among adolescents aged 13–15 years in Saudi Arabia. Additionally, it explored the accessibility of tobacco products among adolescents aged 13–15 years in Saudi Arabia. Moreover, the study examined factors that influence access to tobacco products.

Data sources and participants

The current study is cross sectional with nested case control study, based on data collected via the GYTS questionnaire, which was carried out in Saudi Arabia in 2022. The survey was performed by a work team composed of members from the Ministry of Health (MOH) and the Ministry of Education (MOE). Out of Out of 6983 intermediate school students (grades 1–3). A total of 5610 (80.4%), aged 13–15, from 95 schools and 275 sampled classrooms took part in the study.

Data collection tool

The GYTS is a nationwide representative cross-sectional, school-based questionnaire of students aged 13–15 (corresponding to the 1st to 3rd grades of intermediate schools).

The GYTS, which is part of the Global Tobacco Surveillance System (GTSS), is a global standard for systematically monitoring youth tobacco use (both smoking and smokeless) and tracking key tobacco control indicators.

GYTS employs a standardized core questionnaire, sample design, and data collection protocol, supporting countries in meeting their obligations under the WHO FCTC by generating comparable data across and within nations.

The questionnaire addressed the following topics: prevalence of tobacco use, cessation, SHS, pro- and anti-tobacco advertising and promotion, access to and availability of tobacco products, and knowledge and attitudes regarding tobacco use. To protect data confidentiality, the survey did not require personal information.

Approval was obtained from the MOE and the principals of the selected schools. Parental consent was secured before data collection. Trained personnel from both the Saudi MOH and MOE provided instructions for accurately completing the survey and supervised the process. Students were informed that their participation was voluntary and that they could withdraw at any time or skip any questions they preferred not to answer.

Sampling method

The GYTS employs a two-phase, globally corresponding sampling technique, and schools are chosen with a likelihood proportionate to their enrollment size. The classrooms inside the selected institutions are chosen at random, and all the students in those classes are qualified to complete the questionnaire. The survey uses a standardized core questionnaire supplemented with optional questions to measure and track key tobacco control indicators. The questionnaire is self-administered on scannable, paper-based bubble sheets and is anonymous to ensure confidentiality.

In Saudi Arabia, sampling and weighting are conducted separately for two strata: boys and girls. The national dataset is compiled by merging the data files from all regions. The sampling frame includes all schools with grades 1–3 intermediate levels and 40 or more students.

A two-stage cluster sampling design ensures a representative sample of students.

School level: The first-stage sampling frame included all schools with grades 1 through 3 and an enrollment of

40 or more students. Schools were selected with a probability proportional to their enrollment size.

Class level: The second sampling stage involved systematic equal-probability sampling (with a random start) of classes from each participating school. All classes in the selected schools were included in the sampling frame, and all students in the selected classes were eligible to participate in the survey. Additionally, test–retest results for similar tobacco survey items were used to enhance reliability and reduce bias caused by adolescents' self-reported smoking behaviors and attitudes [24]. In addition to the high response rate (92.3%). The GYTS sample was weighted using a factor applied to each student to adjust for nonresponse "by the school, class, student, and probability of variation of selection either at school or class levels" [25]. This weighting strategy ensured that the survey sample was nationally representative of all Saudi adolescents aged 13–15 years [26]. The data collected was securely stored with access to limited personnel involved in the analysis.

Study variables and measures

The first part of the GYTS collected demographic information, including age, sex, grade level, and weekly personal expenditure, providing foundational data about the respondents. The GYTS core questionnaire was adapted, translated, back-translated, and piloted among students aged 13–15 years to ensure the accuracy and comprehension of the questions.

Descriptive, bivariate, and multivariate analyses were performed. A chi-square ($\chi 2$) test was used to evaluate the associations between participants' characteristics and the use of any tobacco product. The tobacco products used included cigarettes, shisha, waterpipes, heated tobacco, smokeless tobacco, or electronic cigarettes.

Statistical analysis

The data was weighted based on the survey design characteristics. The following weighting formula was used:

$$W = W1 \times W2 \times f1 \times f2 \times f3 \times f4$$

W1 is the inverse of the probability of selecting the school, W2 is the inverse of the probability of selecting the class within the school, f1 is a school-level nonresponse adjustment factor calculated by school size category (small, medium, large), f2 is a class adjustment factor calculated by the school, f3 is a student-level nonresponse adjustment factor calculated by class, and f4 is a poststratification adjustment factor calculated by sex and grade.

The SPSS software package, version 25, was used for secondary analyses of the 2022 GYTS data Chi square test was performed to estimate the associations between smoking status (current versus ever smoking) and variables such as sex, age, amount of pocket money, exposure to SHS at home, and either promotional or antismoking messages, and receiving education about the consequences of tobacco smoking within the past year. All test results were deemed statistically significant at a significance level of ($P \leq 0.05$).

Logistic regression was performed to detect the predictors of being ever or current smoker; odds ratios (OR) were reported with the corresponding 95% confidence intervals (CI). The findings of the OR are considered statistically significant at the $P < 0.05$ level if the 95% CI does not include the one. The predictors are considered as negative predictors if OR > one and considered as protective when OR < one. The variance inflation factor (VIF) Test of multicollinearity used to measure the correlation and strength of correlation between the predictor variables in a regression model. A (VIF) value of 1 was detected indicating that there is no correlation between a given predictor variable and any other predictor variables in the model.

Results

A total of (5610) Saudi male and female intermediate school students (13–15 years age) with a response rate (92.3%) participated in the current study. Table 1 provides an overview of tobacco use, highlighting the significant gender differences ($P < 0.05$). as the study revealed that 8.8% of adolescents (10.0% of boys and 7.5% of girls) were current tobacco users, including shisha and heated tobacco products. Among these, 2.8% smoke cigarettes, 2.8% use shisha, 1.6% smoke heated tobacco products and 3% use e-cigarettes. In contrast, a historical perspective on tobacco usage expresses an increase in the trial attempts of various tobacco products, with 34.7% of males and 28% of females reporting ever using tobacco($P < 0.001$). Deeper analysis of specific forms of tobacco reveals distinct sex differences in cigarette smoking. Currently, 3.4% of male respondents are cigarette smokers, whereas 2.2% of female respondents are cigarette smokers. This difference was statistically significant, with a chi-square value of 8.080 ($P < 0.05$). Further Historical data indicate increased prevalence, with 14.9% of males and 10.8% of females reporting having smoked cigarettes at some point, demonstrating a robust significant disparity ($P < 0.001$). Fortunately, most of the study population falls into the non-tobacco user category, with 90.0% of males and 92.5% of females currently not using tobacco. However, the historical context is less favorable, with only 68.5% of respondents across all sexes reporting that they never used tobacco. Based on a detailed analysis of tobacco use across sex and age groups, Table 2 highlights a clear distinction between current and ever tobacco users. Among current tobacco users (495

Table 1 Prevalence of different types of tobacco use among current and ever tobacco adolescents

Characteristics	Current					Ever				
	Male	Female	χ2	*P* value	Total	Male	Female	χ2	*P* value	Total
	N (% within gender)	N (%within gender)				N (%within gender)	N (%within gender)			N (%)
Any Tobacco product										
Yes	294 (10.0)	201 (7.5)	10.574	< 0.05	495 (8.8)	1020 (34.7)	747 (28.0)	29.059	< 0.001	1767 (31.5)
No	2647 (90.0)	2468 (92.5)			5115 (91.2)	1921 (65.3)	1922 (72.0)			3843 (68.5)
Cigarette smoking										
Yes	101 (3.4)	58 (2.2)	8.080	< 0.05	159 (2.8)	439 (14.9)	288 (10.8)	21.224	< 0.001	727 (13.0)
No	2840 (96.6)	2611 (97.8)			5451 (97.2)	2502 (85.1)	2381 (89.2)			4883 (87.0)
Shisha smoking										
Yes	97 (3.3)	61 (2.3)	5.24	< 0.05	158 (2.8)	461 (15.7)	372 (13.9)	3.339	> 0.05	833 (14.8)
No	2844 (96.7)	2608 (97.7)			5451 (97.2)	2480 (84.3)	2297 (86.1)			4777 (85.2)
Heated tobacco products										
Yes	48 (1.6)	40 (1.5)	0.161	> 0.05	88 (1.6)	262 (8.9)	168 (6.3)	13.509	< 0.001	430 (7.7)
No	2893 (98.4)	2629 (98.5)			5522 (98.4)	2679 (91.1)	2501 (93.7)			5180 (92.3)
E-cigarette smoking										
Yes	176 (6.0)	124 (4.6)	4.952	< 0.05	300 (5.3)	448 (15.2)	343 (12.9)	6.553	< 0.05	791 (14.1)
No	2765 (94.0)	2545 (95.4)			5310 (94.7)	2493 (84.8)	2326 (87.1)			4819 (85.9)
Smokeless tobacco										
Yes	119 (4.0)	69 (2.6)	9.221	< 0.05	188 (3.3)	306 (10.4)	198 (7.4)	15.258	< 0.001	504 (9.0)
No	2822 (96.0)	2600 (97.4)			5422 (69.7)	2635 (89.6)	2471 (39.3)			5106 (91.0)

χ2 Chi square, $P \leq 0.05$ is statistically significant

individuals), the majority were male, with 294 (59.4%) actively using tobacco products compared with 201 (40.6%) who reported no current use. This sex disparity is particularly pronounced among tobacco users. There was a greater proportion of males, with 1020 (57.7%) documented as having engaged in tobacco use at least once, than 747 females (42.3%). Statistical analysis revealed significant differences in tobacco use among individuals aged 13–15 years. For example, 145 students aged 13 years (23.3%) actively used tobacco, while 14 years old accounted for 36.8%, and 168 students aged 15 years (33.9%) were current users. Several factors significantly elevate the prevalence of students' tobacco use, such as weekly average spending less than 30 Saudi Riyals, similar results of SHS exposure at home and school, and parents' smoking with the level of significance ($P < 0.05$). interestingly the participants showed statistically higher smoking levels when they are exposed to messages concerning smoking behavior ($P < 0.001$) and when they undergo orientation sessions at schools about smoking drawbacks on their health ($P < 0.05$).

Table 3 displays social factors related to smoking habits among youth as we find a statistical difference among smokers either current or ever and non-smokers when they receive free samples from a tobacco company representative, Watching T.V show or video or movies where people are smoking ($P < 0.000$). The same results present concerning personal beliefs or emotions and perceptions about smoking among the study participants (e.g., feeling attractive, self-confident or prestigious in social gatherings when smoking) ($P < 0.001$).

Table 4 presents an analysis of cigarette-purchasing habits among male and female smokers. Regarding the sources of cigarette purchases, the results indicate that a greater percentage of male smokers (54.5%) reported purchasing their last cigarettes from stores, shops, street vendors, or the internet than 41.7% of female smokers did. Further analysis revealed that 43.6% of the males bought their last cigarettes from a store, whereas only 25% of the females bought them.

Table 5 shows the predictors of smoking among adolescents in Saudi Arabia. The data indicated a significant difference in smoking rates based on sex, with males showing a notable increase in both current (OR 1.349, 95% CI 1.118–1.627) and ever-smoked categories (OR 1.352, 95% CI 1.207–1.516). These odds suggest that male adolescents are more likely to engage in smoking behaviors than their female counterparts are. The

Table 2 Socio demographic Characteristics of Current & Ever tobacco Users (All products) among adolescents

Characteristics	Current tobacco users all Products (N = 5610)				Ever tobacco users all Products			
	Yes (N = 495) N (%)	No (N = 5115) N (%)	χ2	*P* value	Yes (N = 1767) N (%)	No (N = 3843) N (%)	χ2	*P* value
Gender								
Male	294 (59.4)	2718 (92.4)	10.574	< 0.05	1020 (57.7)	1921 (50.0)	29.059	< 0.001
Female	201 (40.6)	2511 (94.1)			747 (42.3)	1922 (50.0)		
Age								
13	145 (23.3)	1783 (31.8)	12.562	< 0.05	529 (29.9)	1399 (36.4)	35.642	< 0.001
14	182 (36.8)	1958 (34.9)			669 (37.9)	1471(38.3)		
15	168 (33.9)	1374 (24.5)			569 (32.2)	973 (25.3)		
Average Spending								
No/ < 30 SAR	256 (52.2)	2944 (58.1)	6.200	< 0.05	994 (57.0)	2206 (57.8)	0.287	> 0.05
≥ 30 SAR	234 (47.8)	2126 (41.9)			749 (43.0)	1611 (42.2)		
Exposure to SHS								
Yes	305 (61.7)	1948 (38.2)	103.942	< 0.001	938 (53.2)	1315 (34.3)	179.299	< 0.001
No	189 (38.3)	3154 (61.8)			825 (46.8)	2518 (65.7)		
Exposure to SHS At home								
Yes	174 (35.2)	837 (16.4)	107.739	< 0.001	504 (28.6)	507 (13.2)	192.479	< 0.001
No	320 (64.8)	4265 (83.6)			1259 (71.4)	3326 (86.8)		
Students who have one or more parents who smoke								
Yes	144 (29.5)	771 (15.1)	67.512	< 0.001	414 (23.6)	501 (13.1)	97.054	< 0.001
No	344 (70.5)	4333 (84.9)			1343 (76.4)	3334 (86.9)		
Students who saw anyone smoking inside the school building or outside on school property								
Yes	199 (43.6)	1017 (20.3)	131.249	< 0.001	547 (32.3)	669 (17.8)	141.586	< 0.001
No	257 (56.4)	3987 (79.7)			1149 (67.7)	3095 (82.2)		
Exposure to smoking promotion messages								
Yes	341 (68.9)	2338 (45.7)	97.198	< 0.001	1049 (59.4)	1630 (42.4)	139.409	< 0.001
No	154 (31.1)	2777 (54.3)			718 (40.6)	2213 (57.6)		
Exposure to anti-smoking messages (any)								
Yes	355 (71.7)	3238 (63.3)	13.873	< 0.001	1212 (68.6)	2381 (62.0)	23.135	< 0.001
No	140 (28.3)	1877 (36.7)			555 (31.4)	1462 (38.0)		
Students who Taught about the danger of tobacco use during the past 12 months								
Yes	207 (43.8)	1854 (36.9)	8.701	< 0.05	659 (38.2)	1402 (37.2)	0.486	> 0.05
No	266 (56.2)	3171 (63.1)			1068 (61.8)	2369 (62.8)		

χ2 Chi square $P \leq 0.05$ is statistically significant

Table 3 Behavioral beliefs of Current & Ever tobacco Users (All products) among adolescents

Characteristics	Current tobacco users all Products (N = 5610)				Ever tobacco users all Products			
	Yes N (%)	No N (%)	χ2	*P* value	Yes N (%)	No N (%)	χ2	*P* value
Offered a free tobacco product from a tobacco company representative								
Yes	348 (75.7)	256 (5.1)	248.20	< 0.000	231 (13.6)	137 (3.6)	184.14	< 0.000
No	111 (24.3)	4750 (94.9)			1471 (86.4)	3627 (96.4)		
Saw anyone using tobacco on television, videos, or movies								
Yes	195 (62.7)	1853 (47.2)	27.69	< 0.000	708 (56.1)	1340 (45.1)	43.28	< 0.000
No	116 (37.3)	2072 (52.8)			554 (43.9)	1634 (54.9)		
Think that young people who smoke are more attractive								
Yes	132 (27.7)	648 (12.9	79.21	< 0.000	384 (22.2)	396 (10.5)	134.32	< 0.000
No	344 (72.3)	4391 (87.1)			1347 (77.8)	3388 (89.5)		
Students who thought other people's tobacco smoking is harmful to them								
Yes	171 (35.4)	2616 (51.9)	48.128	< 0.001	740 (42.9)	2047 (53.9)	56.836	< 0.001
No	312 (64.6)	2422 (48.1)			983 (57.1)	1751 (46.1)		

χ2 Chi square, $P < 0.05$ is statistically significant,

Table 4 Source of purchasing tobacco products among current adolescent smokers in Saudi Arabia

Characteristics	Gender		χ2	*P* value
	Male (N = 55)	Female (N = 36)		
	NO (%)	NO (%)		
A store, shop, street vendor, or the internet				
Yes	30 (54.5)	15 (41.7)	1.444	> 0.05
No	25 (45.5)	21 (58.3)		
A store				
Yes	24 (43.6)	9 (25.0)	3.270	> 0.05
No	31 (56.4)	27 (75.0)		
A street vendor				
Yes	2 (3.6)	2 (5.6)	0.191	> 0.05
No	53 (96.4)	34 (94.4)		
The internet				
Yes	4 (7.3)	4 (11.1)	0.400	> 0.05
No	51 (92.7)	32 (88.9)		
Someone				
Yes	12 (21.8)	10 (27.8)	0.422	> 0.05
No	43 (78.2)	26 (72.2)		
Were denied purchase of cigarettes				
Yes (not prevented & allowed to sell)	25 (53.2)	12 (46.2)	0.332	> 0.05
No (prevented to sell)	22 (46.8)	14 (53.8)		

χ2 Chi square, $P \leq 0.05$ is statistically significant,

analysis further revealed a clear correlation between age and smoking prevalence. A significant increase in smoking prevalence was observed among adolescents aged 13 and 14 years, with results being significant for current smokers and highly significant for all smokers (CI 0.566–0.756 and 0.684–0.902, respectively). Similarly, exposure to SHS was highly significantly correlated with smoking behavior among participants (CI 1.450- 2.358 among current smokers and CI 1.343–1.797 among ever smokers. Similar results were detected when studying The beliefs of the students towards smoking such as their feeling of socially prestigious, attractive and difficulty to quit.

Discussion

Although school-based survey findings apply only to students who attended school on survey days, the high response rate (92.3%) helps reduce bias caused by nonresponse or absence [27]. The GYTS sample was weighted by a weighting factor that is applied to each pupil to adjust for non-response (by school, class, and student) and variation in the probability of selection at schools and class levels (WHO & CDC, 2014a) [27].

Current study highlights the prevalence of smoking among Saudi adolescents and explores many interacting factors affecting this behavior. It revealed a relatively high prevalence of current (8.8%) and ever using tobacco products using all types (31.5%). Besides that, current and ever users of various tobacco products, including cigarettes, shisha, heated, smokeless tobacco, and e-cigarettes ($P < 0.05$). This aligns with a similar Saudi study conducted by Shubayr et al., (2024) who reported that tobacco usage was prevalent amongst intermediate pupils, with around one-third having tried it and nearly 9% currently using it [14]. It's also consistent with a systematic review conducted in Saudi Arabia by [6] who stated that the prevalence of tobacco smoking among Saudi youth ranged between 2.4% and 39.6%. Although, these studies set alarm to the importance of intense interventions for more strict control measures to decrease the spread of the smoking habit among youth in Saudi Arabia for better adult health and quality of life. Fortunately, they also reflect a reduction in the prevalence of tobacco use among students in Saudi Arabia compared with the findings of Nazir et al. (2019), who analyzed global tobacco use prevalence based on data from the WHO Global Health Observatory (GHO) and reported a prevalence of 15.15% in Saudi Arabia [28]. This aligns with WHO expectations that adolescent tobacco use in Saudi Arabia will decrease from 19.9% in 2000 to 19.2% by 2025 [19]. Encouragingly, this prevalence is lower than the global prevalence, which includes data from 133 countries and shows the highest rates in high-income countries, where 24.76% of males and 19.4% of females use tobacco [28].

Concerning factors affecting adolescent smoking in Saudi Arabia, this work indicates that adolescents with less than 30 Saudi Riyals (SAR) weekly pocket money are more likely to smoke or even try smoking than those who spend 30 SARs or more per week. This is consistent with the findings of [27] who stated that the probability of smoking is negatively related to weekly pocket money where, adolescents who earn more than 50 Egyptian pounds per week are less likely to smoke compared to those who do not receive any weekly pocket money [27]. Similarly, another study findings indicated that youth who have pocket money are more prone to smoking [29]. On the contrary, previous research has suggested that teenagers who spend more money are more likely to smoke [30].

This study revealed that students exposed to SHSs were more likely This finding aligns with a study conducted in Japan, which revealed a strong association between SHS exposure and adolescent smoking [31]. Global evidence also highlights a link between SHS and susceptibility to smoking [32]. Another study suggested that SHS exposure may increase the likelihood of smoking [33]. Besides, boys reported greater exposure to smoking-related influences than girls did, possibly due to social norms and sex-specific exposure risks, which warrants further investigation. Where girls in the region are more likely to stay at home, potentially increasing their exposure to SHS from smoking by parents or adult relatives. In contrast, boys are more likely to spend time in

Table 5 Regression analysis of factors that influence adolescent (13–15 years) smoking in Saudi Arabia (GYTS, 2022)

Characteristics	Current smoking = 381		Ever smoking (1767)	
	OR	95% CI	OR	CI
Gender				
Female	Reference Category			
Male	1.349	1.118–1.627*	1.352	1.207–1.516**
Age				
13	Reference Category			
14	0.674	0. 533–0.851*	0. 654	0.566–0.756**
15	0.768	0.616–0.958*	0.785	0.684–0.902**
Exposure to SHS				
No	Reference Category			
Yes	1.849	1.450–2.358**	1.554	1.343–1.797**
Exposure to SHS At home				
No	Reference Category			
Yes	1.636	1.261–2.122**	1.820	1.525–2.172**
Students who have one or more parents who smoke				
No	Reference Category			
Yes	1.977	1.589–2.459**	1.826	1.573–2.120**
Exposure to smoking promotion messages				
No	Reference Category			
Yes	2.357	1.912–2.905**	1.836	1.629–2.069**
Anti-smoking messages (any)				
No	Reference Category			
Yes	1.319	1.021–1.705*	1.351	1.164–1.568**
Students who saw anyone smoking inside the school building or outside on school property				
No	Reference Category			
Yes	2.899	2.371–3.544**	2.154	1.882–2.466**
Students who Taught about the danger of tobacco use during the past 12 months				
No	Reference Category			
Yes	1.311	1.075–1.598*	1.006	0.826–1.136
Getting tobacco products from public source (Store-Shop-Vendor-Internet)				
No	Reference Category			
Yes	0.889	0.519–1.522	1.173	0.497–2.767
Current tobacco smokers who were not prevented from buying cigarettes because of their age				
No	Reference Category			
Yes	0.372	0.227–0.610*	0.578	0.270–1.238*
Offered a free tobacco product from a tobacco company representative				
No	Reference Category			
Yes	0.301	0.202–0.449**	0.429	0.312–0.590**
Saw anyone using tobacco on television, videos, or movies				
No	Reference Category			
Yes	0.519	0.404–0.667**	0.652	0.568–0.747**
Think that young people who smoke are more attractive				
No	Reference Category			
Yes	0.576	0.418–0.794**	0.509	0.416–0.624**
Students who thought other people's tobacco smoking is harmful to them				
No	Reference Category			
Yes	0.564	0.461–0.690**	0.698	0.620–0.786**
Students who thought smoking tobacco helps people feel more comfortable at celebrations, parties, and social gatherings				
No	Reference Category			
Yes	2.376	1.950–2.894**	1.996	1.755–2.271**
Students who thought it is difficult to quit once someone starts smoking tobacco				
No	Reference Category			
Yes	1.093	0.834–1.432	0.899	0.758–1.065

OR: Odds Ratio, CI: Confidence Interval, $P < 0.05$ is statistically significant, *statistically significant results, **Highly significant

public settings where smoking is common, which leads to higher SHS exposure in many countries hence, the Adolescent's growing interest in trying smoking [20]. Another study by Kheirallah and his team (2016) revealed that the growth of Waterpipe Tobacco Smoking (WTS) in the Arab countries began in the early 1990s and become a socially accepted habit among adolescents in these societies when the water pipe became a focal point of social gatherings [34]. Therefore, there is a need for interventions specifically designed for male-dominated spaces. Fortunately, it was stated that greater knowledge of the harmful effects of SHS reduces susceptibility to smoking [29].

In our study, a surprisingly finding is that adolescents exposed to either smoking promotion or anti-smoking messages experienced higher prevalence of smoking than those not exposed to such messages ($P<0.05$). This aligns with the findings of Chassin et al. (2014), who analyzed GYTS data from 168 countries and reported that smoking-prone adolescents were more likely to receive anti-smoking messages [30]. Similarly, Mishu et al. (2021) demonstrated that exposure of youth to anti-tobacco media messages did not prevent tobacco use [35]. In contrast, [36] reported that adolescents exposed to tobacco advertising for various products were more likely to start using them, including smoking cigarettes, even if they only recalled marketing for distinct products. On the other hand, this study revealed that adolescents taught at school about the hazards and health effects of smoking were less likely to be current or former smokers. This finding agrees with that of [35], who reported that adolescents informed at school about the adverse consequences of tobacco use were less likely to use smokeless tobacco [35]. Similarly, Logo et al. (2021) reported that students who received education about the harmful effects of tobacco were less likely to use these products [37]. We could explain these findings by the fact that the age of the current study participants (13–15 years) is characterized by being a developing stage of adolescence, which often involves a great desire for more knowledge and more experience, leading to the willingness of young people to experiment new things and to imitate adults.

Current work also revealed that there's a significant relationship between youth exposure to media such as advertisements, videos and or films and their smoking behavior, Similar findings reported by [38] who stated that a strong link between tobacco-related media material and smoking vulnerability. Likewise, we found that the beliefs of feeling attractive or self-confident in social gatherings increases the adolescents' vulnerability to smoke. This is confirmed by [38] who found that Adolescents thought that they perceive greater social benefits of smoking. With respect to the tobacco accessibility characteristics of the study participants, there was no statistically significant difference between the methods of obtaining tobacco products through various public means (stores, shops, vendors, or the internet) among adolescents currently smoking cigarettes. This is because the Saudi anti-smoking Law [23] prevents youth under the age of 18 years from purchasing tobacco products. This underscores the need for more strict enforcement of age-related sales regulations and greater oversight of public sources to reduce access to and limit tobacco use among adolescents. This is critical because one of the primary objectives of tobacco control measures is to prevent children and adolescents from smoking during their developmental years, a period when they are particularly vulnerable to nicotine dependence [39]. These results are further supported by the findings of Cruz et al. (2019), who noted that adolescents using the internet are more likely to recall marketing for various tobacco products, increasing their risk of initiating tobacco use and cigarette smoking [36].

Logistic regression of the available GYTS data revealed that being a male and the age of 14/ or 15 years are predictors for adolescents in Saudi Arabia to try, currently use tobacco products. This finding is logic because males tend to be more active, have broader social networks, and are more socially engaged than females in the Arab societies. This coincides with the results of [27] who reported that in Egypt, the risk of smoking rises with age, with males being more likely to experiment and continue than females. Additionally, as adolescents grow older, their social circle expands, increasing their likelihood of encountering tobacco products. Moreover, exposure to SHS at home or school is a predictor of being a current smoker or even trying smoking, as they may perceive this behavior as normative. This aligns with Ba-break et al. (2021), who reported that school environments and surrounding settings significantly influence youth smoking habits. For example, observing teachers smoking at school was associated with a greater likelihood of current or former smokers among adolescents [27]. Similarly, O'Loughlin and colleagues revealed that Adolescents are more likely to start smoking when they are exposed to it in their social context [40].

On the same side, Monshi et al. (2019, 2021) reported that the socioeconomic development experienced in Gulf Corporation Council (GCC) countries, including Saudi Arabia, in recent decades has profoundly influenced smoking practices and SHS exposure [41, 42]. In this context, educating students about the health hazards and disadvantages of smoking has been shown to reduce their willingness to try tobacco products [19]. Adolescents' perceptions of the harmfulness of smoking and social context also played crucial roles among the current study participants. This study highlights that adolescents' beliefs that smoking tobacco helps them feel attractive

and more comfortable at celebrations, parties, and social gatherings are predictors of smoking maintenance. Similar findings were reported in an Egyptian study conducted by Ba-Break et al. (2021), who reported that the belief in feeling comfortable when using tobacco during celebrations and social occasions increases the probability of tobacco use among adolescents [27]. This highlights social patterns and areas for intervention. However, students who believed that it would be difficult to quit once someone started smoking tobacco did not have a greater probability of using tobacco products. The above findings can inform targeted interventions to reduce adolescent smoking rates in Saudi Arabia.

Conclusion

In summary, the findings of the current work highlight the more prevalent smoking habits among male adolescents than females. It also, illuminate that low pocket money, SHS exposure, personal feeling self-confident are common predictors of smoking among youth in Saudi Arabia. Such results underscore the need for sex-specific interventions designed to effectively promote tobacco cessation in both sexes. As we continue to investigate these dynamics, the implications for health policy and education may become more profound, potentially offering innovative strategies to address tobacco-related health disparities. This study highlights a critical concern: the proportion of young individuals experimenting with tobacco reflects broader societal challenges, underscoring the need for effective educational programs. These statistically significant findings serve as important alerts for stakeholders committed to reducing tobacco use. The prevalence of smoking among adolescents remains a pressing public health issue globally. Therefore, understanding the demographic and environmental factors influencing smoking behaviors is essential for developing prevention strategies and promoting healthier lifestyles among youth.

Together, these insights drive actions and dialog surrounding tobacco use, particularly among different sexes and age groups. They highlight the importance of strengthening prevention measures.

Recommendations

In future studies we recommend using objective measures to overcome the limitation of the subjective nature of the GYTS Survey studies. Besides that, we suggest longitudinal study for better determination of the predictors of adolescent smoking. In addition, further in-depth research projects are needed to investigate the effect of friends' smoking (peer pressure) among youth in Saudi Arabia such as studying the effect of peer selection (smoker and non smoker) on the smoking behavior among adolescents' groups. Also, wider introduction of comprehensive school smoking policies such as ban of smoking in schools are required to reduce smoking rates among this critical age category. Also, combined efforts between the MOH and the MOE, schools and healthcare providers may excute educational campaigns tailored to this targeted populations. Moreover, strict control of smoking products advertisements in different channels of social media at the same time of arranging antismoking education campaigns through this social media channels. Hence shifting the current situation from concerning to hopeful and encouraging healthy habits for future generations. Furthermore, it might be appropriate to reflect and document Saudi experience in tobacco control.

Strengths

As a national survey, the GYTS data accurately reflect national trends, and a reliable representation of tobacco use patterns among the participant age groups in Saudi Arabia. Moreover, the high response rate among the study participants is a notable strength, enhancing the reliability of the survey results.

Limitations

This study has several limitations. First, the GYTS relies on self-reported data, which are inherently susceptible to social desirability bias and underreporting, particularly with respect to sensitive behaviors such as tobacco use. However, research indicates that self-reports of smoking habits are plausible [38]. Furthermore, the GYTS is a school-based survey, potentially excluding youths not enrolled in school or those unwilling to participate in school-based research. Although this may result in the underrepresentation of certain demographic groups at greater risk for tobacco use. However, the high response rate of the study participants overcomes this issue. Additionally, adolescents may provide socially desirable responses, especially when sensitive topics are addressed. Another important limitation that there are no sufficient responses to study the peer effect that acts as an important item in studying the smoking behavior among youth. As Previous research suggested that smoking beginning was more likely if teenagers belonged to a smoking group; however, smokers were not more likely to stop if they belonged to a non-smoking group and peer group smoking homogeneity is influenced by both selection and influence. Hence further detailed research should be directed only to study different aspects of peer effect.

Recognizing these limitations is essential for accurately interpreting the study findings and guiding future research and public health initiatives.

Abbreviations

GYTS	Global Youth Tobacco Survey
SHS	Second-hand smoke
WHO	World Health Organization

CDC	Centers for Diseases Prevention and Control
ATPs	Alternative tobacco products
FCTC	Framework Convention on Tobacco Control
MPOWER	Monitor, Protect, Offer, Warn, Enforce, Raise
GCC	Gulf Corporation Council
GTSS	Global Tobacco Surveillance System
e-cigarettes	Electronic cigarettes
MOH	Ministry of Health
MOE	Ministry of Education
GTSS	Global Tobacco Surveillance System
χ2	Chi-square
SAR	Saudi Riyals
GHO	Global Health Observatory
WTS	Waterpipe Tobacco Smoking

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Author contributions
MME, AAE and MAS conceived and designed the study and wrote the manuscript. SSM, AHAZ and EDED: conducted literature searches for the first draft of the manuscript, conceptualized and supervised the data analysis, and analyzed the data. SKA and MMA: designed the study and supervised data collection. ASA and MSA reviewed and approved the final draft before submission. All the authors read and approved the final version of the manuscript.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Ethical approval was obtained from the Central Institutional Review Board of the Ministry of Health (MOH), Saudi Arabia, IRB number 23-106M (Date: 17 December 2023). The GYTS survey was conducted in accordance with the ethical guidelines laid down in the Declaration of Helsinki. All participants' parents/guardians provided written informed consent before taking part in the survey.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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